

Issue Management Framework

Formalizing issues and problem management as an organizational competency

Goals

- ❑ Providing a centralized capability to manage issues, risks, questions, risks, and tasks.
- ❑ Provide a baseline model to feed into Demand Management
- ❑ Provide a holistic capability for driving change and identifying opportunities across operational areas as well as transitional activities (associated with change management & acquisition management).

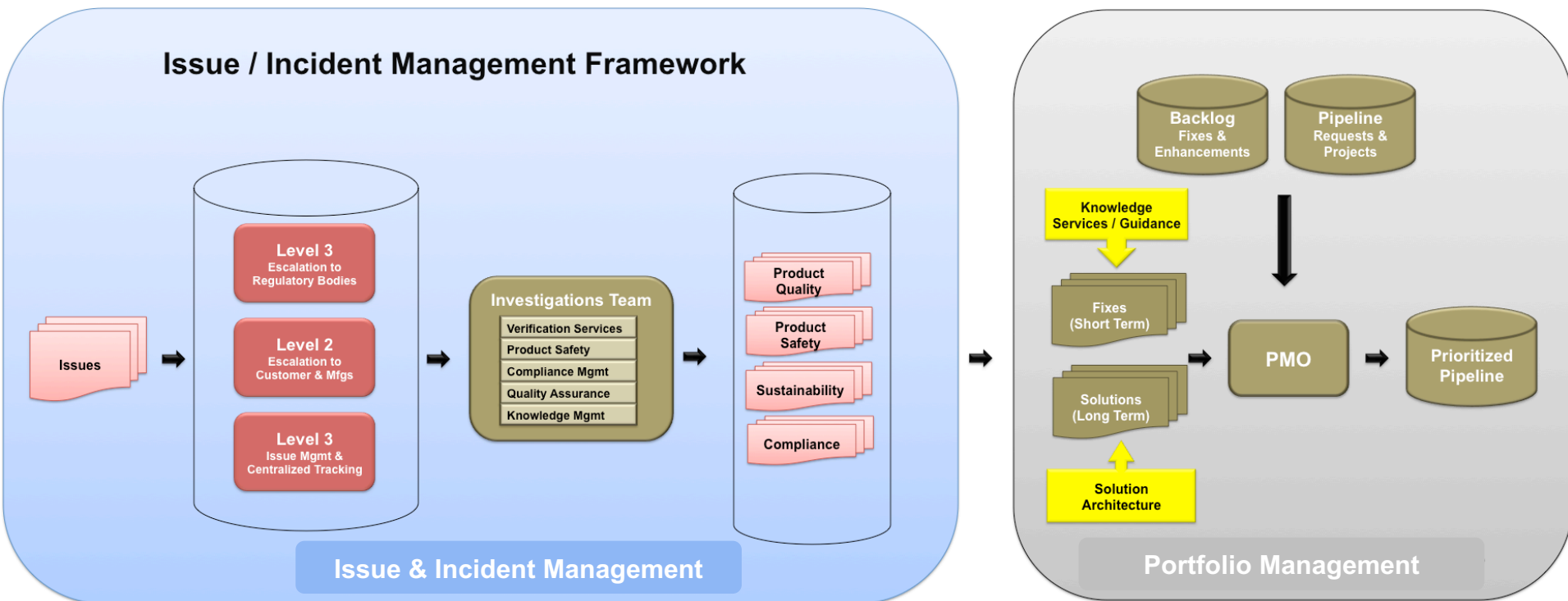
Scope & Opportunity

- ❑ Creating an issue management solution using a common repository (virtualized footprint) for capturing and cataloguing issues. Issues are triaged by to determine causal factors for issue and follow up information is provided for resolution via workflow.
- ❑ Immediate need is to provide guidance for a data model, process, and integration points which need to be considered for the short-term pilot and while ensuring these can scale into a larger program which can be used across the organization and its customers.
- ❑ The bigger opportunity involves developing a Continuous Improvement Model that can be deployed as an enterprise capability using a federated set of tools and processes.

Issue Management w/in Continuous Improvement

- ❑ Issue Management is the main feed used to drive change in organizations. Issues are a superset of problems, requests, defects, and gaps with customer needs or compliance (risk).
- ❑ Recommendations for changing current state are derived from examining issues, which are then bundled into projects or tasked out to support and/or operations staff to implement.
- ❑ The goal for continuous improvement is to reduce the likelihood of issues recurring (i.e. don't repeat mistakes) thus reducing the backlog of reactionary work. Proactive work should be the bulk of requests in the pipeline once a steady-state level of maturity is achieved.
- ❑ In order to achieve this level of maturity, all issues need to be captured and tracked using a consistent data model to conduct patterned-based mining of root causes and/or opportunities.

Continuous Improvement Framework



- ❑ The same framework can be used to manage issues or incidents
- ❑ Opportunities for improvement can flow through several areas. Guidance for continuous improvement (w/in the organization and across the value chain) can be combined with Knowledge Services to improve the organization's value proposition.

Proposal for an Enterprise Solution

Centralized Issue Management

- ❑ The proposal involves creating a centralized repository using a standard “logical” model (from which a physical schema can be derived).
- ❑ All existing (and proposed) applications which are (will be) capturing issues, risks, tasks, defects, questions, and requests will be modified to write directly to a centralized issue repository.
- ❑ In some cases, data may need to be staged but the goal is to create a singular container for issues.
- ❑ Our standard BI tools can be applied against the repository and dashboards can be built for operational monitoring & for project teams managing change.
- ❑ Issues related to data quality and/or security may need to be reviewed by a separate team with forensic skills to identify root cause and provide guidance on remediation to governance teams.

Types of Issues (Classification)

❑ Problems – anomalies which occur in production (products/services)

- Severity is a further classification which identifies impact to the organization, systems, and its customers
- Incidents are a further classification which occur when a customer is impacted. Note: if a customer is injured or death occurs, these incidents become reportable to one or more regulatory authorities

❑ Defects – concerns / anomalies which occur in across the lifecycle development (products or software)

- These need to be tracked with problems since some (many) of these defects may not be addressed during lifecycle development but remain “dormant” as potential risk

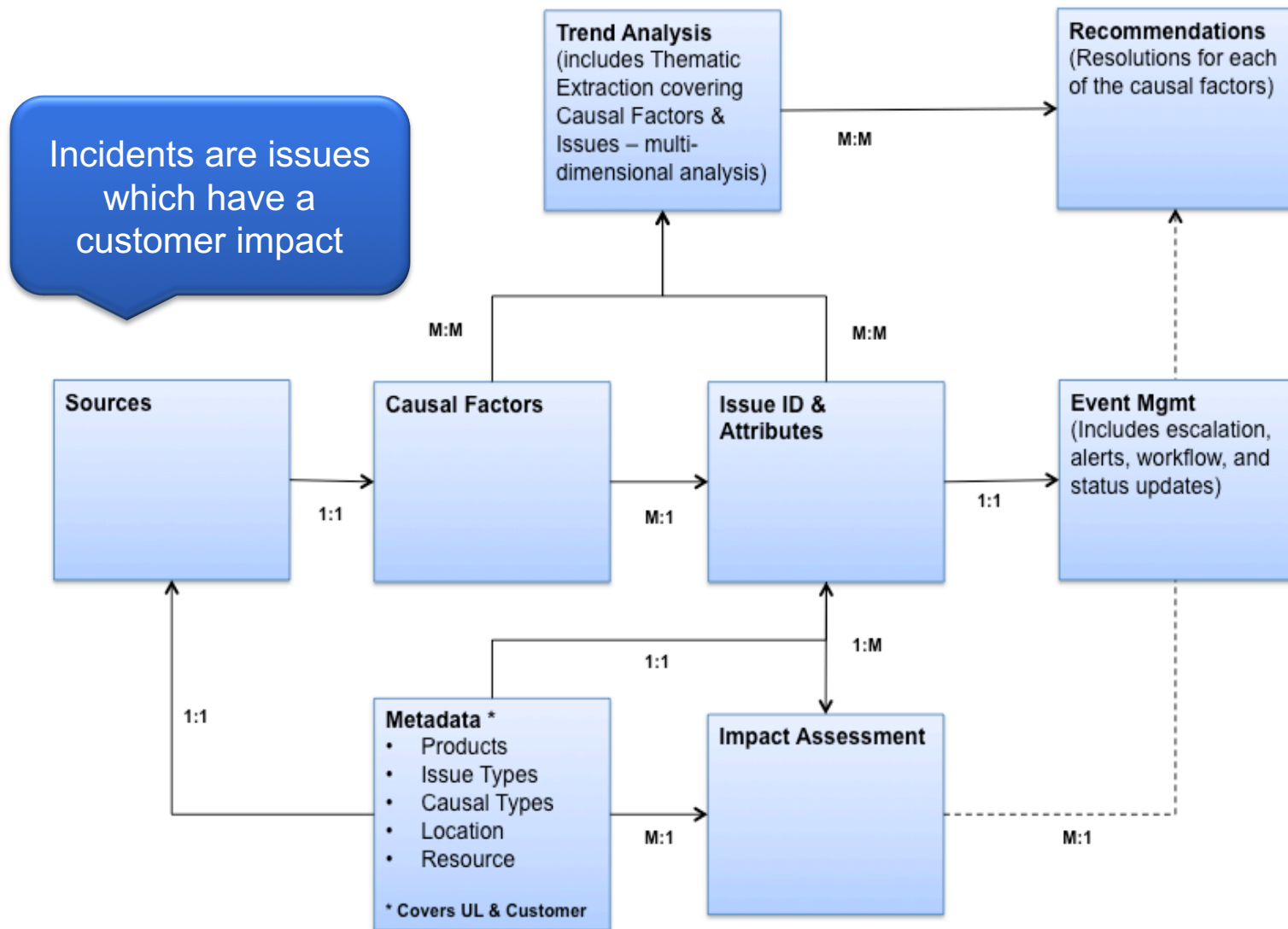
❑ Requests – internal / external requests for change

- Customers requesting changes in products / services may be an indicator of either problems or defects. Typically, these are missed requirements or opportunities that were never discovered during conceptualization
- Internal staff may identify opportunities for improvement (using formal methods like LEAN, 6Sigma, or Kaizen) or informally through experience and simple observation

Types of Issues (Classification)

- ❑ **Risks – may or may not be related to compliance gaps, but they should be tracked as an issue since issues may be linked to risks which have not been mitigated. Covers the following 6 areas:**
 - **Strategic**
 - Flaws in the strategy, business plan, or business model can sink good & great companies
 - **Operational**
 - Exposures with Quality, Operational Excellence, & Productivity (efficiency & effectiveness)
 - Includes Customer Focus, Quote to Cash, Supply Chain & Logistics, PPM & Lifecycle Mgmt
 - **Financial**
 - Includes risky investments and/or budget management (balancing short & long term)
 - **Compliance**
 - Regulatory risk is when you are not compliant with the laws & statutes (local or federal)
 - Security risk spans across the business & regulatory
 - Standards risk is when you are not following consistent (recognized) practices
 - **Technical**
 - Architectural debt (technical, technology)
 - **Data**
 - Covers the lifecycle of data, information, and knowledge to ensure quality & integrity

HL Data Model for Issue Management



Extending Issue Management to Customers

❑ Problems – anomalies which occur in production (products/services)

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❑ Same approach is used to track issues across their lifecycles to drive continuous improvement for products and services

- Extending Issue Management involves using the same framework (data model, processes, classification, event management). The rules & metadata will vary based on contextual needs for each customer.
- The goal is to have a consistent framework in place across the organization. Built properly, this can be extended as a service offering for Business Unit's to support data mining for trends and opportunities for driving continuous improvement in a more holistic manner.
- The same framework can be extended as a service offering (B2C) for customers

Defining the Scope for Issue Management

❑ WHAT

- Security is becoming increasingly important as it relates to products & services. Issues can impact any of the cornerstones and can span across all of the Business Unit's. By leveraging these areas of compliance for tracking issues provides entry into driving continuous improvement horizontally (orthogonal to the functional & Business Unit silos).

❑ WHERE

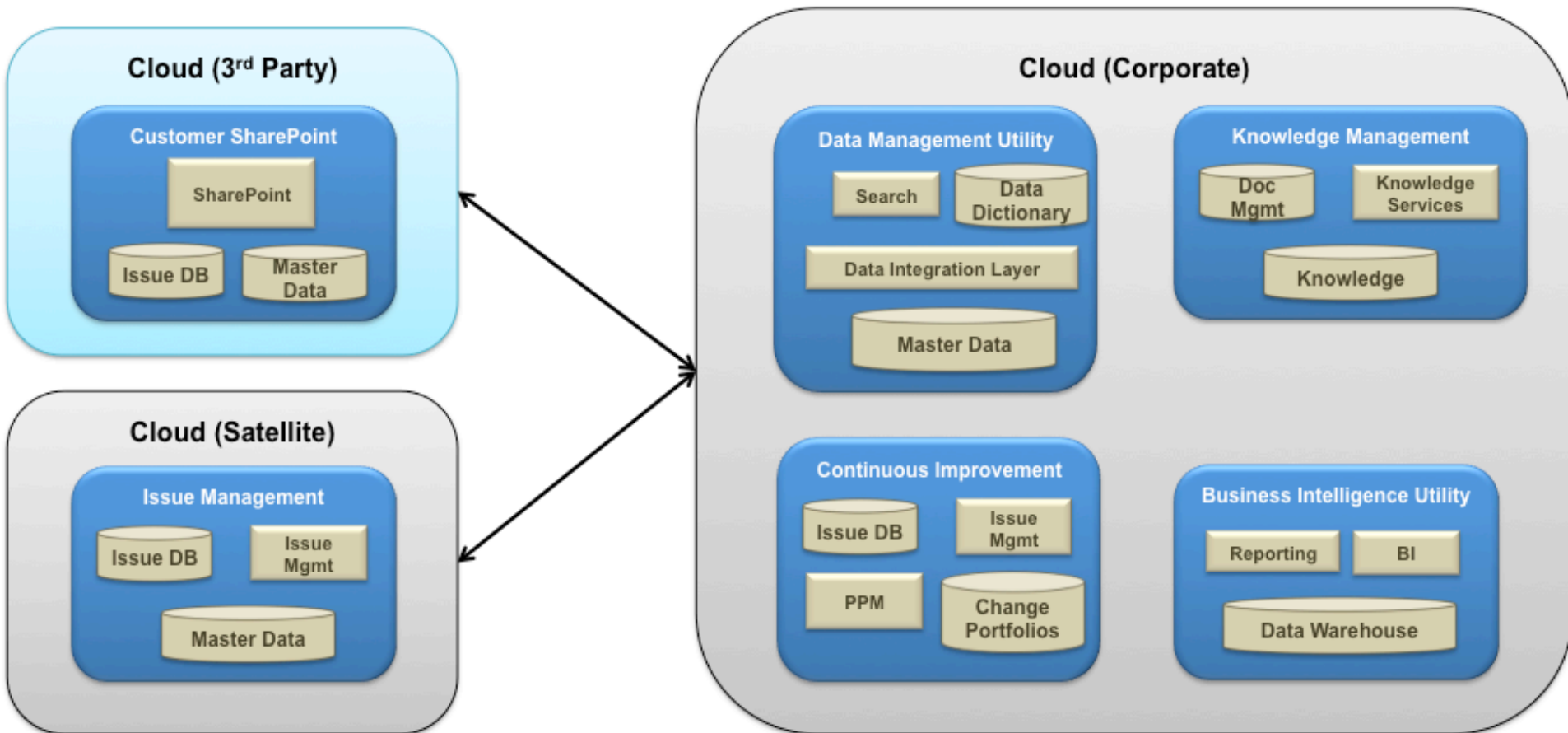
- Issues can occur at any point in our operations as well as service delivery for our customers. Organizations focused on moving service delivery further up the value creation (PLM) as well as value delivery (SCM), so the opportunity for tracking issues across the product lifecycle and driving continuous improvement for customers is huge.
- We also want to track issues that impact our operational excellence across the various functions and Business Unit's since these can impact service delivery as well as effectiveness.

❑ WHO, WHY, & HOW

- Once issues have been captured, they need to be analyzed for causal factors to identify why and how the issue arose and which resources (who) were involved. This requires disciplined techniques combined with tools for performing forensic analysis.

High Level Architecture

Cloud-Based Service Model



Overview: Investigations Team

- ❑ **May involve members from the following teams/areas:**
 - Quality Assurance
 - R&D and Product Safety
 - Product Development
 - Customer Service

- ❑ **The same members may be involved in coming up with guidance and/or recommendations for Customer's (or their Suppliers) on ways to prevent issues (including incidents) from recurring.**
 - Allows the organization to draw upon its knowledge services covering product design, material science, sustainability, and compliance (rules and tolerances derived from regulations, standards, & best practice).

Overview: Investigations Team

❑ Investigations Team

- Identify research opportunities based on analysis of existing or potential risk involving data (internal and/or external).
- Conduct research to identify causal factors for data issues
- Collaborate with associate teams in the DG Operations Team (typically) to identify options and propose a recommendation for remediating the data issue (short term to address the immediate problem and long term to prevent a reoccurrence of the problem).

❑ Some tools for Investigations Team may include:

- Fishbone analysis can identify numerous sources for causality.
- Data Mining / Mapping tools to align causality across prior / existing issues to reveal patterns and lead to better solutions.

Overview: Incident Management

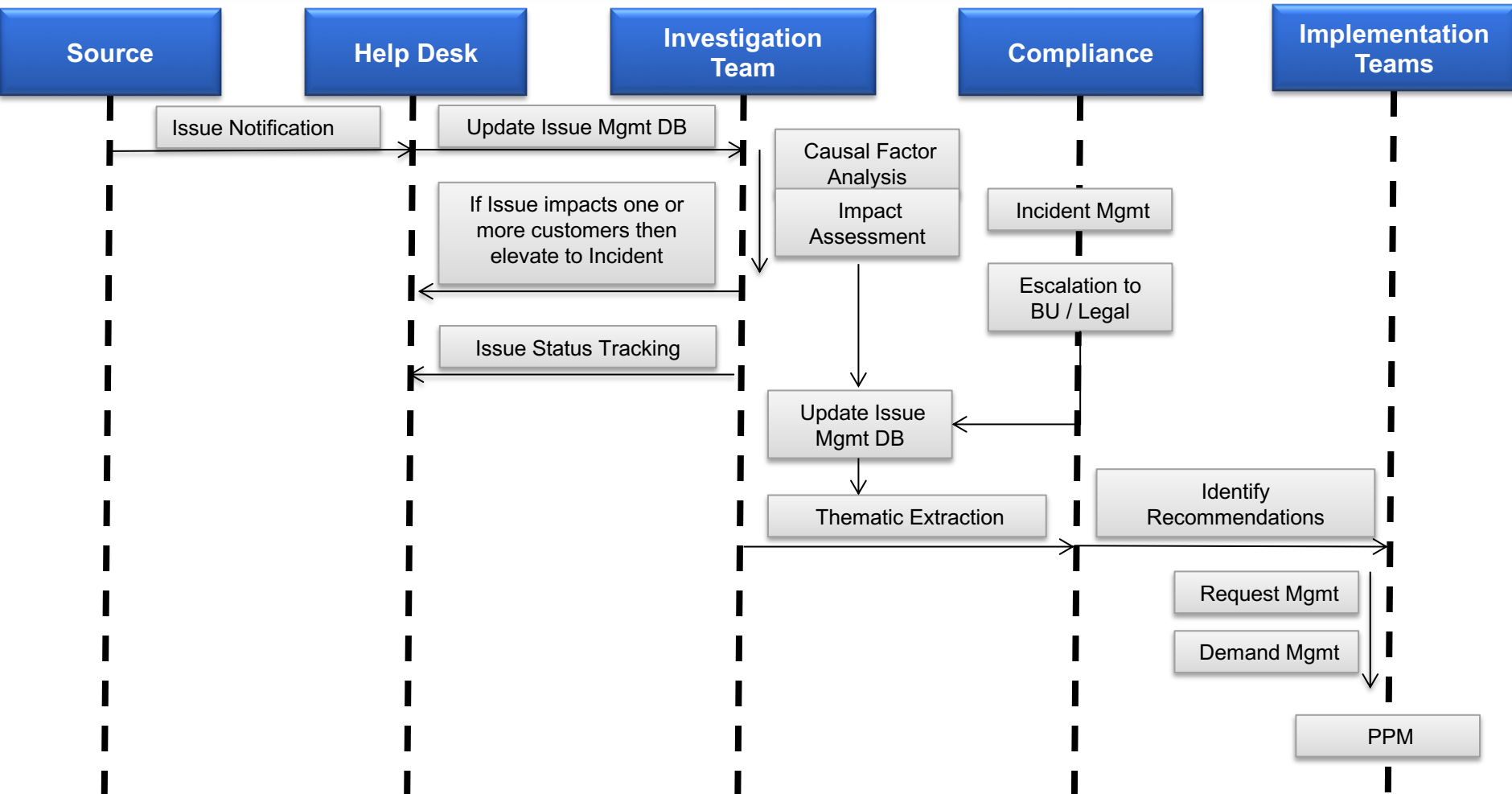
- ❑ **Once an issue is identified as an incident (has customer impact), the next step is to assess the impact on the customer:**
 - Level 3 – Customer identified a defect, but no harm was reported. May or may not be a product defect.
 - Level 2 – Customer identified a defect and one or more customers may have been harmed (no death). Product defect was verified.
 - Level 1 – Customer identified a defect AND one or more customers were harmed with at least one death confirmed due to product safety

- ❑ **In any case where a product defect is confirmed, a recall may be recommended and communications to regulatory authorities may be required.**

Overview: Causal Factor Analysis

- ❑ The Investigations Team provides expertise with techniques for identifying root cause(s) associated with data issues. These techniques can be found in the discipline for Causal Factor Analysis. Root cause analysis is used by organizations who strive to achieve highly reliable operational capabilities.
- ❑ The goal is to identify the reasons WHY an event occurred that results in an issue (e.g., an internal problem or an external problem that impacts customers), or an unexplained anomaly (e.g., an event that was not anticipated, but may lead to an issue).

Sequence Diagram for Issue Management



Operational Architecture

❑ Processes (Core Processes)

- Add Issue
- Update Issue
- Investigate Issue
- Impact Analysis from Issue
- Causal Factor Analysis (CFA) for Issue
- Trending Analysis using CFA output
- Thematic Extraction across Root Causes & Patterns (trends)
- Knowledge Mining from Resources (compliance, materials, testing, products)
- Ideation for Addressing Root Causes (prioritized list of recommendations)
- Communication of CFA to Customer (and/or Suppliers = Mfgs)
- Escalate Issues (based on Severity Level to regulatory bodies)
- Audit of all above processes
- Recursive Feedback of Issue Management to drive Continuous Improvement

Operational Architecture

❑ Process Models, Metrics, & Controls

- Create SIPOCs for each process
- Model all Integration Points (and for interoperability)
- Identify Metrics & Reports to monitor controls, performance, customer satisfaction, and compliance needs are met (for the organization and its Customers)
- Ensure Audit trails and reporting are in place
- Ensure Security concerns are met (all parties)
- Ensure Error Management need are met
- Ensure Key Metrics are identified
- Ensure Reporting for Quality, Performance, Compliance, and SLAs are in place and operational

Application Architecture

- ❑ Issues should be stored in a consistent data model across the organization since all processes are aligned to enable value delivery to customers.
- ❑ The data model should be extensible to support the lifecycle management around issues as they flow into resolution for continuous improvement.
- ❑ Traceability for issues should be transparent across all Business Units and functional areas with appropriate releasability to manage exposure and privacy (esp. when they concern and/or affect customers)
- ❑ Formalization of RCA* (forensic analysis to identify causal factors) should be introduced as a discipline that can be extended to customers as part of a value-added service.

*** RCA = Root Cause Analysis**

Application Architecture

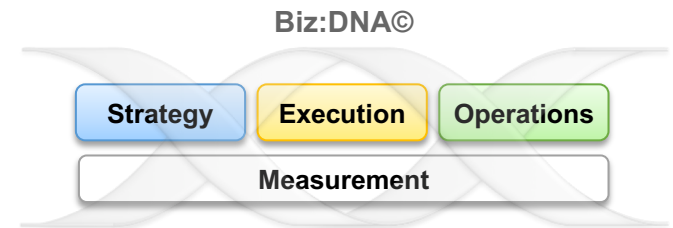
- ❑ Knowledge Services can provide a natural feed into RCA and can augment ideation to identify recommendations for addressing root causes
- ❑ Issue capture (via the Help Desk, Customer Service, or via Systems) should feed into a consistent data model for triage, escalation, and resolution.
- ❑ The issue management framework should be separately managed (albeit using federated sets of data and/or tools) but share a common integration layer, which can feed into Continuous Improvement Frameworks (within the organization or across the value chains where it participates).
- ❑ A common metadata model should be used to feed in consistent dimensions for attribution of issues (initially, and throughout their lifecycle). Typical dimensions include Customer, Product, Location, and Resources.

Application Architecture

- ❑ Applications back-ends will be modified (slightly) to re-direct the data access (manipulation & search) to a set of Data Services (part of a Data Management Utility = DMU)
- ❑ Application front-ends will be modified to ensure they are using a standard set of data fields (w/ pull-downs populated using consistent metadata)
- ❑ Data Profiling, Data Matching, and Data Cleansing routines should be applied within the Integration Layer to ensure all data feeds are consistent with Data Quality Standards / Policies.
- ❑ Security over all data will be managed via the DMU via provisioned filtering applied to data access requests (initially) and to data manipulation events (eventually). The solution should leverage technical capabilities (with sufficient abstraction as this is subject to further evolution & maturity).

Thank You

Measure what matters
Create and deliver the right value
Build and nurture a winning culture
Proactively manage change & complexity



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